

Particulate Matter Monitoring & Health Effects

July 2026 | Entry window 1–31 July 2026 | 157 records pooled

157

RECORDS POOLED
(10 SUBTOPICS)

62

EFFECT ESTIMATES
EXTRACTED

23

SENSING /
INSTRUMENTATION

49

RECORDS FROM CHINA
(31% OF CORPUS)

8

ULTRAFINE /
NANOSCALE

Signal of the month

July was the month the field stopped treating $PM_{2.5}$ mass as the exposure of record. Three independent literatures — ultrafine number, chemical constituents, and unmodelled condensable mass — each put a number on what bulk mass misses.

Lelieveld et al. (*Cardiovasc Res*) built a 1 km global ultrafine-particle field from Earth observation and machine learning, found urban annual means of $10,000\text{--}30,000 \text{ particles cm}^{-3}$, and paired it with a pooled cohort hazard ratio to estimate **1.99 (0.81–3.89) million excess deaths per year** — roughly 5% of non-communicable-disease mortality, about half cardiovascular, with **91% occurring in urban and suburban environments**. They put an annual limit of $5,000 \text{ cm}^{-3}$ at a **45% reduction** in that excess. UFPs are regulated almost nowhere and measured by no mass monitor. Su et al. (*Environ Sci Technol*) supplied the second number: industrial *condensable* PM, absent from every operational source-apportionment model, contributes a median **3.6% (95% CI 0.9–14.5%)** of urban $PM_{2.5}$ in four Chinese cities and **9.5% in winter**, and its omission inflates apparent secondary inorganic aerosol by **5.6–16.6%**. Easton et al. (*Environ Int*) supplied the mechanism: cruise-ship ultrafine PM, enriched in vanadium, nickel and cobalt, was markedly more pro-inflammatory than same-port PM of other origin *and* suppressed the innate antiviral response, increasing epithelial permissiveness to rhinovirus-16 and SARS-CoV-2 — an effect reproduced by vanadium alone. Read together with Zhu et al. (29 Jul issue), where $PM_{2.5}$ was the *null* term in its own six-pollutant time-series, the month's verdict is unambiguous: **a mass-only monitor is now measuring the wrong quantity for a growing fraction of the health signal.**

What changed in July

- **Sensing was the single largest subtopic of the month (23/157, 15%), and that inverts the picture the daily issues gave.** Across the five published daily issues of July the sensing count read **2, 2, 6, 1 and 0** (27–31 July), 11 records in total; the last two issues flagged the low counts as a suspected harvest defect. The month confirms it: instrumentation work was being published throughout July and the narrow daily windows recovered a small fraction of it. No conclusion about the sensing literature should be drawn from any single daily issue in this series.
- **Low-cost sensors were validated in July with unusual honesty about where they fail.** Gharibzadeh et al. needed *seasonal* multivariable calibration (humidity, temperature, PM_{10}) to reach R^2 0.62–0.93 in Tehran, and reported persistent underestimation before it. Ali et al. found accuracy collapsing with particle size — $R^2 \approx 0.90\text{--}0.95$ for PM_{10} but **0.28–0.68 for PM_{10}** — and, critically, that neither the PMS7003 nor the SPS30 captured the dominant sub- $0.30 \mu\text{m}$ fraction at all. Coefficient of variation exceeded 43% in the coarse bins. The message for network design is that proprietary size-bin conversions should not be trusted above $1 \mu\text{m}$.
- **Sensor-plus-satellite fusion produced the month's most striking performance number, and it deserves scepticism.** Jung et al. reported a 10-fold CV R^2 of **0.98 (RMSE $1.23 \mu\text{g}/\text{m}^3$)** for a 250 m daily $PM_{2.5}$ field over Taiwan built from 16,795 low-cost sensors plus MAIAC AOD. That is far above the 0.82 (Singh et al., North-East India) and 0.41–0.80 (Yu et al., Shenzhen sidewalks) obtained elsewhere in the same month. With 16,795 sensors as both predictor and validation target, random 10-fold splits leak spatially; the rolling-origin and spatial folds are the numbers to read, not the headline.
- **Exposure misclassification moved from caveat to quantified bias.** Zhang et al. (*Environ Int*) showed nearest-monitor assignment is *systematically higher* than both scaled-matched estimates and true personal exposure from outdoor sources, for every pollutant tested. Nyoni et al. (31 Jul issue) showed road-geometry proxies inverting in sign between African countries. Kos Sinhala et al. showed in-home black carbon driven by open fireplaces and wood stoves rather than ambient concentration. Three independent designs, one conclusion: the exposure variable in most published epidemiology is not the exposure that occurred.
- **Reproductive and developmental evidence consolidated around vulnerability, not average effect.** Sieber et al. found $PM_{2.5}$ associated with **6% (95% CI 0–11%) higher odds of term low birth weight per $10 \mu\text{g}/\text{m}^3$** across 1.4 million Spanish births, but maternal education

dominated it — primary versus tertiary education carried **OR 1.76 (1.71–1.82)**. Meeting the WHO $5\mu\text{g}/\text{m}^3$ guideline would have prevented 2,032 (95% CI –74 to 3,900) cases over nine years; universal tertiary education would have prevented far more. That comparison is the uncomfortable and important finding.

- **Two well-powered nulls are worth as much as the positives.** Sun et al. found **no** adverse association between $\text{PM}_{2.5}$ and FEV_1 decline in US adults with cystic fibrosis ($+0.002$ z-score per $3\mu\text{g}/\text{m}^3$; 95% CI $-0.001, 0.005$) at a mean exposure of $9.1\mu\text{g}/\text{m}^3$. Tien-Smith et al. found no clear evidence for $\text{PM}_{2.5}$ and multiple sclerosis (1.01; 0.77–1.32) or motor neuron disease (1.07; 0.86–1.33), while confirming Parkinson's disease risk in Asia (1.19; 1.01–1.41). Neither is a negative result about air pollution; both are results about where the exposure-response function is flat.

Corpus analytics — July 2026

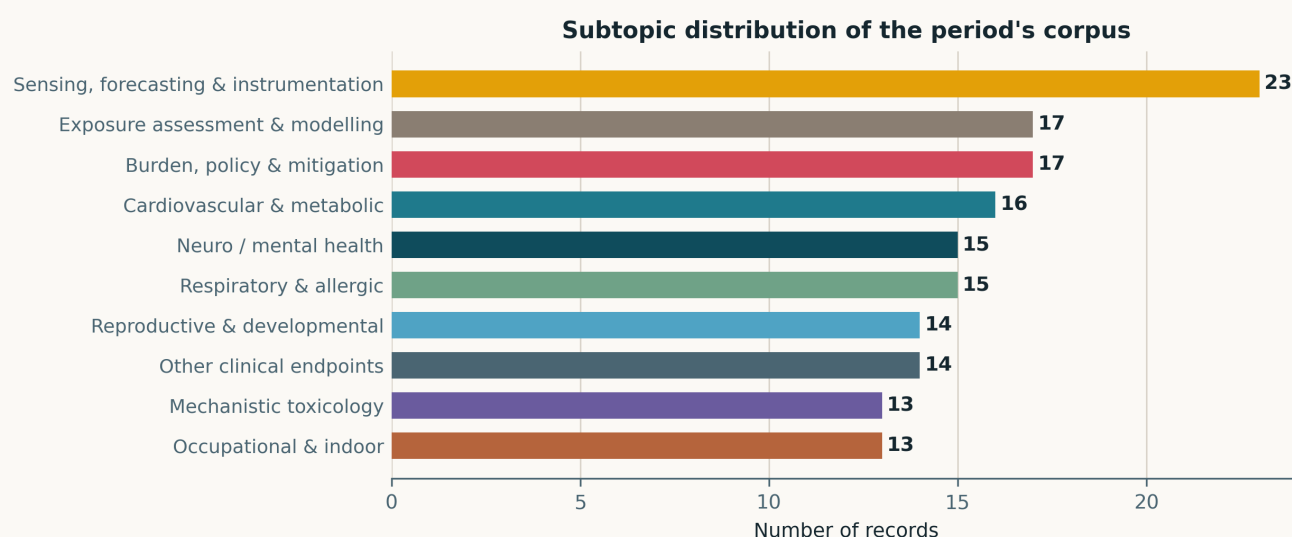


Fig. Subtopic distribution across all 157 pooled records. All ten subtopics are populated — no daily issue in July achieved that. Sensing, forecasting & instrumentation leads at 23 records against 11 across all five published daily issues combined (2, 2, 6, 1, 0 on 27–31 July). Assignment is single-label by dominant contribution.

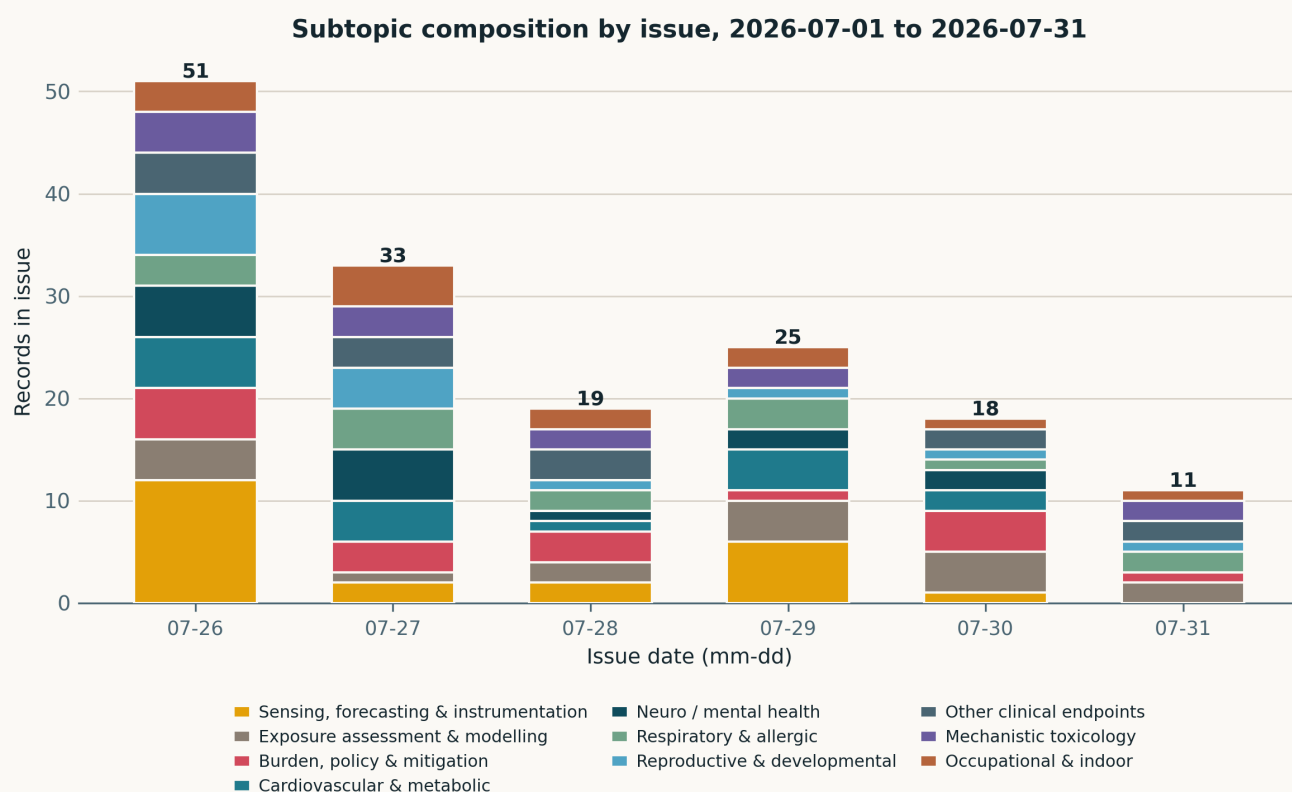


Fig. Records per issue date by subtopic. **Read this as a harvest diagnostic, not as a time series.** The 26 July bar is the retrospective backfill for 1–26 July harvested in a single batch on 1 August; it is not a day's output. Only the 27–31 July bars are genuine daily windows. The panel is included because the contrast is the point: five narrow daily windows recovered 106 records while one month-wide query recovered 51 more than the dailies never saw.

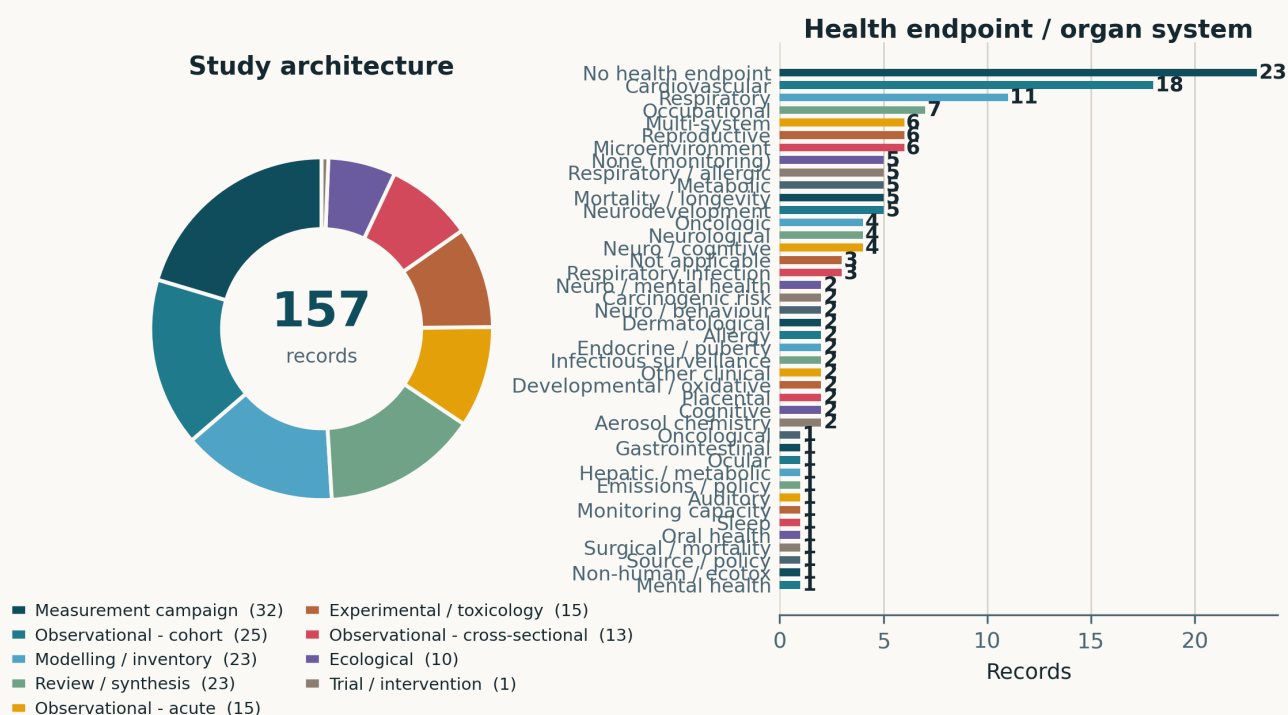


Fig. Left — study architecture over the month. Measurement campaigns are the largest class (32), ahead of observational cohorts (25), modelling/inventory (23) and reviews (23). Only one trial or intervention record appeared in the entire month, which is the standing weakness of this evidence base: almost nothing here is experimental in humans. Right — health endpoint distribution.

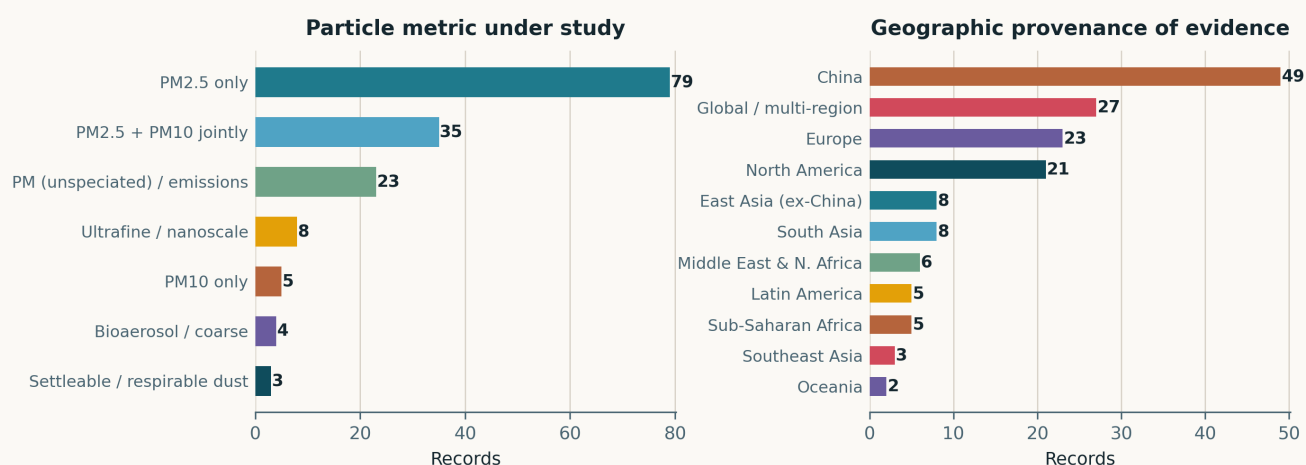


Fig. Left — particle metric. PM_{2.5} alone in 79 records and PM_{2.5} with PM₁₀ in 35; **ultrafine or nanoscale metrics appear in only 8 records (5%)** despite carrying the month's largest single mortality estimate. That mismatch between where the burden is claimed and where the measurement effort goes is the clearest instrumentation gap in this corpus. Right — geography. China supplies 49 records (31%), Europe 23, North America 21; sub-Saharan Africa supplies 5 and Latin America 5.

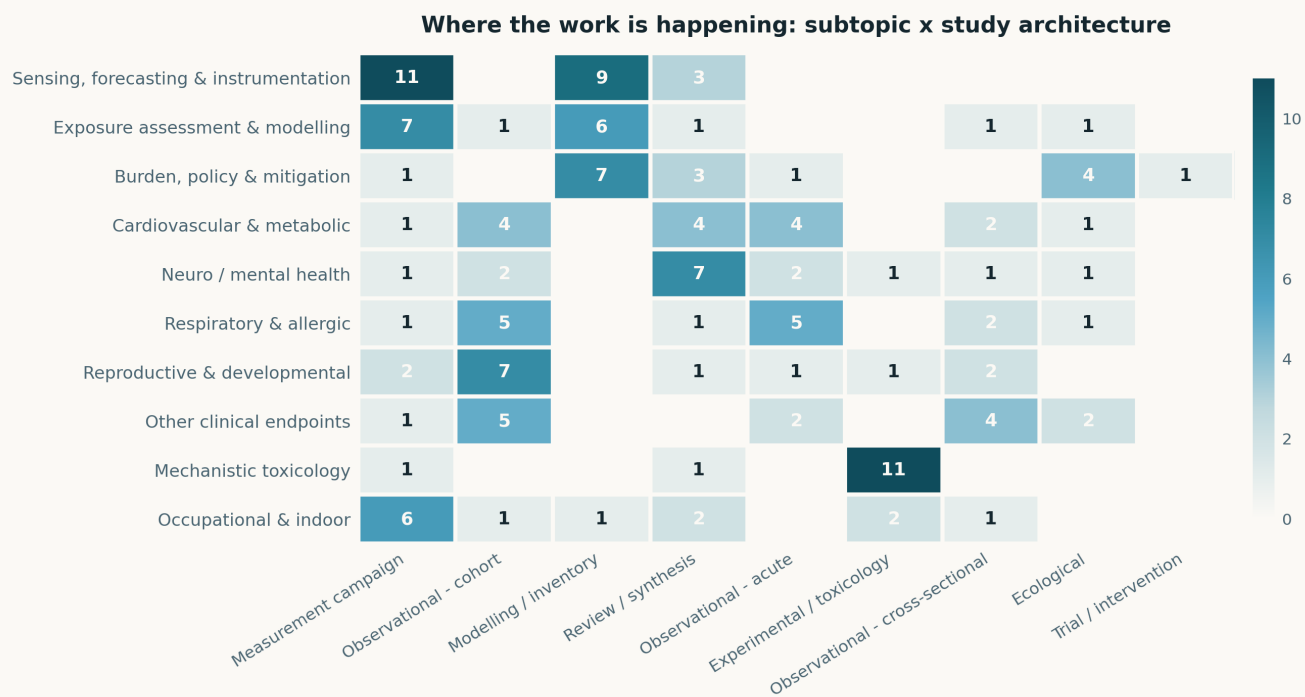


Fig. Subtopic × study-architecture cross-tabulation over the month. The dense Sensing/measurement-campaign and Burden/modelling cells are expected. The informative absence is the near-empty trial column across every health subtopic.

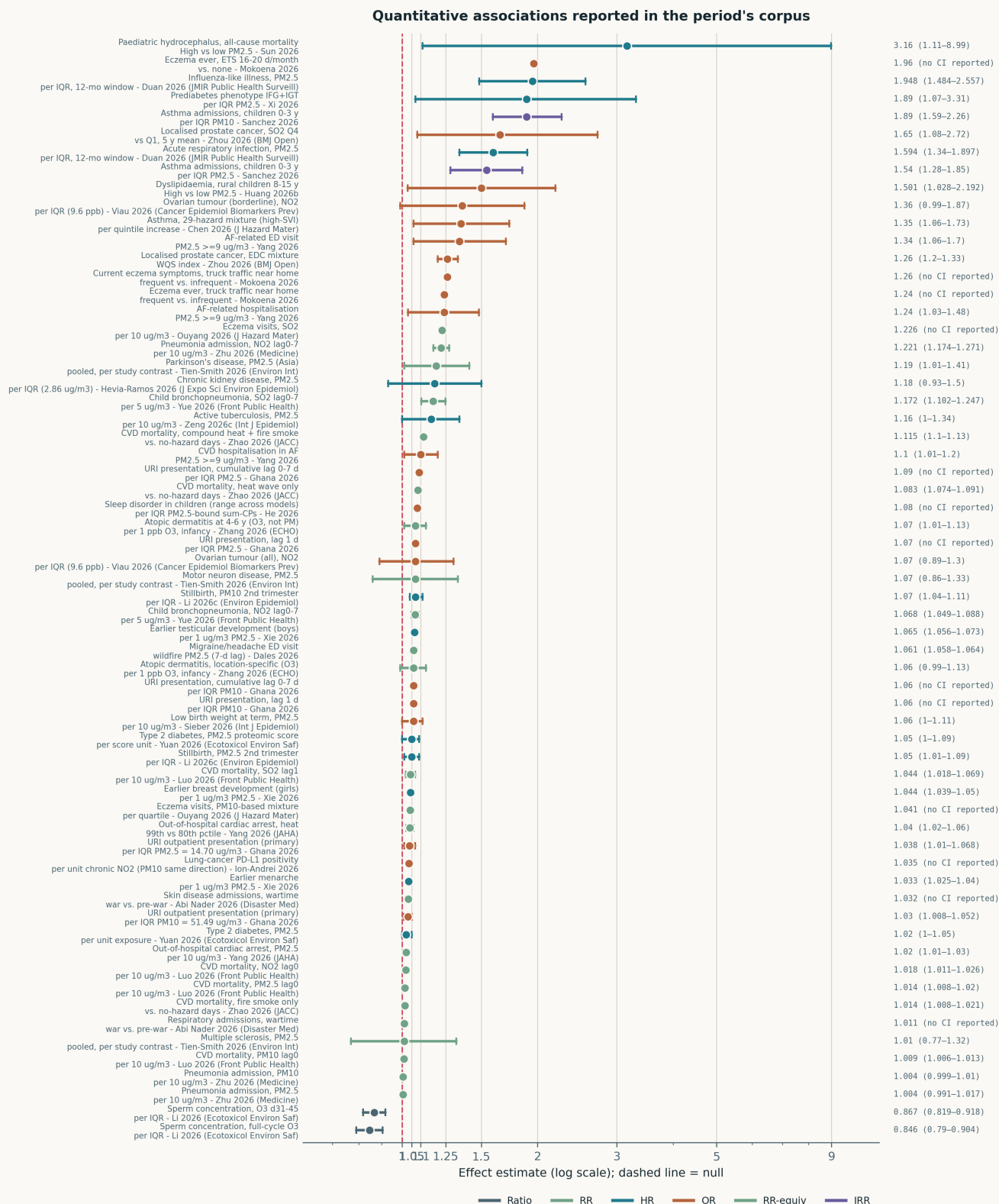


Fig. All 62 quantitative effect estimates extracted across July on a common ratio axis. **This is not a meta-analysis and no pooled summary is drawn.** Exposure contrasts are not commensurable — per-IQR, per-5, per-10 $\mu\text{g}/\text{m}^3$, quartile and quintile contrasts, and mixture indices all appear — and the outcome scale mixes RR, OR, aOR and HR across different diseases. **Heterogeneity is therefore not estimable and none is reported;** the one place a pooled figure would be legitimate is Tien-Smith et al.'s within-paper neurodegenerative meta-analysis, which is reproduced here as three separate points rather than absorbed. The visible structure is that per-10 $\mu\text{g}/\text{m}^3$ PM_{2.5} estimates cluster tightly just above unity (1.02–1.16) while the large ratios belong to per-IQR contrasts, mixture indices and gaseous co-pollutants.

Life-course windows addressed this period (records may span several stages)



Fig. Life-course windows addressed across the month. Counts are non-exclusive and cover only records with an identifiable human life-course window; *in vitro*, *ex-vivo*, animal and pure-instrumentation records are excluded. Adolescence remains the thinnest window in the corpus, as it was in every individual daily issue.

Emerging, fading, and recurring

Emerging. Three themes appeared or firmed up materially this month. (i) **Ultrafine particles as a regulatory object** — Lelieveld et al. and a paired *Cardiovasc Res* commentary, Easton et al.'s ship-source toxicology, Williams et al.'s hepatic xanthine-oxidoreductase mechanism, Zeng et al.'s microglial circRNA work, and Olivares Lopez et al.'s finding that particle-number simulations are strongly sensitive to emission-inventory choice all converge in a single month, driven by the new EU Ambient Air Quality Directive's particle-number monitoring requirement. (ii) **Compound exposures** — PM_{2.5} with heat is now a design rather than a sensitivity analysis (Yang et al., 610,613 OHCAs; Zhang et al., 5.07 million adults). (iii) **Omics as the mediation layer** — proteomic signatures for type 2 diabetes (49,134 UK Biobank participants), gut-microbiome mediation of cognition, cord-blood extracellular-vesicle miRNA, and placental mtDNA copy number and telomere length.

Fading. Nothing can honestly be called fading on one month of data, and the backfill/daily imbalance described in the provenance box would manufacture false declines if pressed. The single defensible observation is that **single-pollutant, single-endpoint time-series studies now read as the low-information tail** of this corpus: several appeared, most were single-city and single-hospital, and the ones that reported a PM term alongside gases generally found PM the weakest term in its own model.

Recurring groups and venues. No author cluster recurs meaningfully at first-author level — the apparent repeats (Li 5, Zhang 4, Wang 4, Chen 4) are surname frequency, not the same investigators, and should not be reported as a cluster. At institutional level, Chinese public-health schools (Zhengzhou, Sichuan, Fudan, CUHK) and the Barcelona/BiSC and Danish/Nordic cohort infrastructures each contribute repeatedly. Venue concentration is modest: *Toxics* (7), *J Environ Sci (China)* (6), *Environ Pollut* (5), *Ecotoxicol Environ Saf* (5), *Environ Sci Technol* (4), *J Hazard Mater* (4). The month's highest-standing placements were *Eur Heart J*, *J Am Coll Cardiol*, *Cardiovasc Res*, *Environ Health Perspect* (3 records), *Environ Int* (3) and *Int J Epidemiol* (2). Citation counts on records days to weeks old are uninformative and were not used to rank anything here.

Trial registry watch

ClinicalTrials.gov returned **one** PM or air-pollution record registered or updated inside the monitored portion of the window: **NCT05160948**, “Assessing Changes in Knowledge, Behavior, and Attitude Regarding Air Pollution After Receiving Personal Air Monitoring Results” (Cincinnati Children’s Hospital Medical Center; observational; $n = 105$; completed). No endpoint analysis was run because the study is observational rather than interventional. It is nonetheless the single most on-topic registry item this watch has seen: it tests whether *returning personal monitor data to the participant* changes behaviour, which is the assumption every citizen-science and personal-sensing deployment rests on and almost none evaluates.

Two records in the peer-reviewed corpus function as the interventional evidence the registry did not supply. González-Colom et al. (*BMJ Open Respir Res*, NCT06421402) monitored household air quality with low-cost sensors in 205 adults with COPD, asthma or bronchiectasis for two months: **51.7% of homes had at least one pollutant in an at-risk category, and the burden was overwhelmingly PM_{2.5}-driven (40.1% of dwellings at risk)**, with active or passive smoking strongly associated ($p < 0.001$). **No relationship was found with hospitalisations or ED visits** — the honest null that a two-month screening window cannot overturn. Bolden et al. reviewed indoor wildfire-smoke mitigation and concluded that portable-air-cleaner efficacy is real but too dependent on building envelope and occupant behaviour to generalise across the national housing stock. **The interventional evidence base for personal and household PM reduction remains thin, and July did not thicken it.**

Open gaps and read on direction

Gaps in the literature

(i) **Ultrafine measurement does not exist at network scale.** Eight records used an ultrafine metric; one of them attributes two million annual deaths to it. There is no low-cost particle-number instrument in this corpus with a published field-calibration comparable to what Gharibzadeh et al. did for PM_{2.5} mass. (ii) **Almost no human intervention studies.** One trial-class record in 157. Every causal claim in this month’s corpus rests on observational designs with modelled exposure. (iii) **Sub-0.3 μm is invisible to the deployed sensor base.** Ali et al. state this directly for the two most widely deployed OPC modules; it is also where the ultrafine burden lives. (iv) **Geographic imbalance is severe.** 49 records from China against 5 from sub-Saharan Africa, in a month whose most methodologically important exposure paper (Nyoni et al.) is precisely about African settings where the standard proxies fail.

Gaps this pipeline creates

(i) **No date-windowed channel into *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*** PubMed does not index them, the Consensus connector has no entry-date filter, and OpenAlex date filtering now returns 403/429. Every instrumentation record in this issue reached the corpus through a biomedically-indexed journal, so the pure atmospheric-measurement literature is *still* absent even at 23 sensing records. The fix remains a Crossref by-journal feed keyed on created for AMT (1867-8548) and ACP (1680-7324). (ii) **Narrow daily windows systematically under-recover.** Demonstrated quantitatively this month: the sensing subtopic read 2, 2, 6, 1 and 0 across the five published daily issues of July (27–31 July) — 11 records in total — against 23 for the month, and the month’s figure is itself a floor. (iii) **The 1–26 July backfill is classification-depth only.** Effect estimates were extracted for that batch only where the abstract reported an interval; a genuine daily cadence would have captured more.

Read on direction. The field is moving from *mass* to *composition and number*, and the instrumentation base has not moved with it. July produced a two-million-death annual estimate for a pollutant metric that no regulatory network measures and no validated low-cost sensor reports; simultaneously it produced careful evidence that the deployed optical sensor base cannot see below 0.3 μm and degrades badly above 1 μm . That gap is the most fundable problem visible in this corpus, and it is an instrumentation problem before it is an epidemiological one. **The falsifiable prediction: within the next two monthly issues, expect at least one field-calibration study of a low-cost particle-number or black-carbon sensor against a CPC or aethalometer reference, framed explicitly against the new EU particle-number monitoring requirement.** If that does not appear by the September issue, the measurement community is lagging the epidemiology by more than the usual cycle, and the right inference is that the bottleneck is hardware cost rather than research attention.

Pooled corpus register — July 2026

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Calderon-Garcidueñas	Abnormal eye movements reflect early cortical and volumetric brain changes in PM _{2.5} -exposed urban youth	<i>J Alzheimers Dis</i>	Cross-sectional imaging	PM _{2.5}	Mexico
Cattarinussi	Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories	<i>medRxiv</i>	Prospective cohort	PM _{2.5}	USA
Dales	Wildfire-sourced PM _{2.5} and ED visits for migraine and other primary headache	<i>JAMA Netw Open</i>	Case-crossover	PM _{2.5} (wild-fire)	Canada
Ghassabian et al.	Prenatal air pollution exposure and autism spectrum disorder in the ECHO Consortium	<i>Environ Health Perspect</i>	Prospective cohort	PM _{2.5} + NO ₂ + O ₃	USA
Glover	Umbrella meta-review of urban environmental exposures and cognitive health	<i>Age Ageing</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Goossens	In utero air pollution exposure and pre-/postnatal brain development	<i>Neurosci Appl</i>	Narrative review	PM _{2.5} / PM ₁₀ / PM _{coarse}	Global
Kabir	Neighbourhood socioeconomic inequalities in mental health: moderation by environmental attributes	<i>Health Place</i>	Scoping review	PM (unspecified) / emissions	Global
Lu	Subchronic PM _{2.5} exposure exacerbates chronic stress-induced depression-like pathology in mice	<i>Environ Res</i>	Murine inhalation exposure	PM _{2.5}	China
Ma	PM _{2.5} -induced neurotoxicity and depression: exposure risks and mechanisms	<i>iScience</i>	Narrative review	PM _{2.5} + components	Global
Ma et al.	Impacts of contemporaneous air pollution exposure on cognitive performance in Kenya	<i>J Health Econ</i>	Panel study	PM _{2.5}	Kenya
Pan	Maternal PM exposure and offspring neurodevelopment: effects and mechanisms	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Narrative review	PM (bulk)	China
Qi et al.	The role of gut microbiota in the association between air pollution and cognitive function in older adults	<i>Environ Health Perspect</i>	Repeated-measures biomarker	PM _{2.5} + O ₃	China
Sarkar	Particulate-matter-induced neurotoxicity: neuroinflammation and organelle dysfunction	<i>J Appl Toxicol</i>	Narrative review	PM _{2.5} + PM ₁₀ jointly	Global
Tien-Smith et al.	Long-term outdoor air pollution and incidence of Parkinson's disease, multiple sclerosis and motor neuron disease: a systematic review and meta-analysis	<i>Environ Int</i>	Umbrella review / meta-analysis	PM _{2.5} + PM ₁₀ + NO ₂	Global
Wang et al.	The evolving clinical landscape of PM _{2.5} -attributable hemorrhagic stroke in China, 1990-2050	<i>Neurol Sci</i>	GBD secondary analysis	PM _{2.5}	China
Cardiovascular & metabolic					
Belay	Long-term bushfire-related PM _{2.5} and coronary plaque burden in asymptomatic adults	<i>Eur J Prev Cardiol</i>	Prospective cohort	PM _{2.5} (bush-fire)	Australia
Chen	Long-term particulate matter exposure and coronary inflammation by pericoronary fat attenuation index	<i>Environ Pollut</i>	Retrospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Chrysoshoou	Preventing cardiovascular disease within the built and living environment: an expert opinion	<i>Eur J Prev Cardiol</i>	Narrative review	PM (unspecified)	EU-27
Huang	PM _{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Cross-sectional (multistage)	PM _{2.5}	China
Kazmierska	Air pollution and cardiovascular disease: review of current data	<i>Przegl Epidemiol</i>	Narrative review	PM _{2.5} / PM ₁₀	Global
Lelieveld et al.	Air quality and health implications of exposure to ultrafine particle pollution	<i>Cardiovasc Res</i>	Umbrella review / meta-analysis	Ultrafine (particle number)	Global
Li et al.	Sequential mediating roles of MMP-12 and DNA methylation in air pollution-induced atherosclerotic cardiovascular disease	<i>Ecotoxical Environ Saf</i>	Repeated-measures biomarker	PM _{2.5} + PM ₁₀ + PM ₁ components	China
Luo	Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning	<i>Front Public Health</i>	Time-series	PM _{2.5} + PM ₁₀	China
Paolisso et al.	Micro- and nano-plastics in the coronary circulation and air pollution exposure in ischaemic heart disease presentation	<i>Eur Heart J</i>	Retrospective clinical series	PM _{2.5} (+ micro/nanoplastics)	Italy

First author	Focus	Journal	Design	Metric	Region
Rusconi	Long-term ultrafine particle exposure 2010-2019 in high-income countries and cardiovascular burden	<i>GeoHealth</i>	Ecological panel	Ultrafine particles	Global
Xi	Air pollution, built environment and temperature vs. prediabetes phenotypes	<i>Diabetologia</i>	Prospective cohort	PM _{2.5}	Germany
Yang	Short-term PM _{2.5} and hospitalisation/ED use in atrial fibrillation	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} (wildfire-influenced)	USA
Yang et al.	Joint effects of heat and fine particulate matter on out-of-hospital cardiac arrest in Japan	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5}	Japan
Zhang et al.c	Association of joint exposure to PM _{2.5} and extreme heat with cardiovascular disease: a population-based cohort study	<i>iScience</i>	Prospective cohort	PM _{2.5}	China
Zhao	Compound exposure to extreme heat and substantial landscape fire-sourced air pollution and cardiovascular mortality	<i>J Am Coll Cardiol</i>	Case-crossover	PM _{2.5} (fire-sourced)	Global
Zhou	Air pollution and hypertension: umbrella review of systematic reviews and meta-analyses	<i>iScience</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀	Global
Reproductive & developmental					
Chen	Prenatal air pollution, fetal growth trajectory phenotypes and child neurodevelopment	<i>J Environ Sci (China)</i>	Prospective cohort	PM _{2.5} / PM ₁₀	China
Gomez-Herrera et al.	Prenatal air pollution exposure and placental biomarkers of oxidative stress and ageing: the BiSC cohort	<i>Int J Hyg Environ Health</i>	Repeated-measures biomarker	PM _{2.5} + BC + NO ₂	Spain
Hu et al.	Prenatal exposure to ambient air pollution and fetal growth trajectories in Wuhan, China	<i>Environ Health Perspect</i>	Prospective cohort	PM _{2.5} + gases	China
Huang	Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures)	PM _{2.5} / PM ₁₀	China
Kahroba et al.	Cord-blood black carbon particle burden and a C19MC extracellular-vesicle miRNA signature enriched for neurodevelopmental pathways	<i>Environ Res</i>	Repeated-measures biomarker	Black carbon (cord blood)	UK
Li	Ambient air pollutant exposure and semen quality: a refined multi-window panel study	<i>Ecotoxical Environ Saf</i>	Panel study	PM _{2.5} + PM ₁₀	China
Li et al.c	Maternal ambient air pollution exposure and risk of stillbirth in Georgia, USA	<i>Environ Epidemiol</i>	Case-control	PM _{2.5} + PM ₁₀ + 8 pollutants	USA
Lin	Maternal particulate matter exposure during pregnancy and offspring liver function	<i>Environ Pollut</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Liu	Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models	<i>Int J Mol Sci</i>	In vitro (cell lines)	PM _{2.5}	Netherlands
Peterson	Ambient air pollution, PAHs and pubertal development: critical review	<i>Curr Environ Health Rep</i>	Critical review	PM _{2.5} components	Global
Sezer et al.	Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate	<i>Birth Defects Res</i>	Case-control	PM _{2.5} + PM ₁₀ + O ₃	Turkiye
Sieber et al.	The impact of improved air quality on low birth weight and health inequalities in Spain, 2010-2018	<i>Int J Epidemiol</i>	Registry cohort + geospatial	PM _{2.5} + NO ₂	Spain
Xie	Long-term PM _{2.5} and timing of pubertal development in boys and girls	<i>Environ Res</i>	Prospective cohort	PM _{2.5}	China
Yin et al.	PM _{2.5} -dominant disruption of male reproductive hormone homeostasis: fine versus coarse particulate matter	<i>Ecotoxical Environ Saf</i>	Prospective cohort	PM _{2.5} + PM _{2.5-10} + PM ₁₀	China
Respiratory & allergic					
Bayar Muluk	Climate change and otolaryngology: airway disease impacts and emission reduction	<i>Int J Biometeorol</i>	Narrative review	PM _{2.5}	Global
Chaiwong	PM _{2.5} and systemic inflammatory biomarkers in elderly COPD patients versus healthy elderly	<i>Int J Mol Sci</i>	Panel study	PM _{2.5}	Thailand
Duan et al.	Long-term exposure to PM _{2.5} and ozone and the risk of acute respiratory infections: community-based prospective cohort	<i>JMIR Public Health Surveill</i>	Prospective cohort	PM _{2.5} + O ₃	China
Ghana AIRCLARE	Short-term ambient PM _{2.5} /PM ₁₀ and acute respiratory morbidity in Ghana: time-stratified case-crossover pilot	<i>Research Square (preprint)</i>	Case-crossover	PM _{2.5} / PM ₁₀	Ghana
Li	Human genetic variation, airway microbiome, environmental exposure and respiratory health	<i>mSystems</i>	Prospective cohort	PM _{2.5} only	China

First author	Focus	Journal	Design	Metric	Region
Liao	Short-term air pollution, respiratory symptoms and lung function in asthma	<i>Med Gas Res</i>	Panel study	PM _{2.5} / PM ₁₀	China
Mokoena	Socio-economic conditions, air pollution sources and eczema symptoms among teenagers in Soshanguve	<i>Public Health Chall</i>	Cross-sectional (multistage)	PM (unspecified) / emissions	South Africa
Odimba et al.	Sex-specific associations between ambient air pollution exposure thresholds and lung function in middle-aged and older Canadians	<i>Inhal Toxicol</i>	Cross-sectional (multistage)	PM _{2.5} + NO ₂ + O ₃	Canada
Sanchez de Prada	Ambient PM and asthma-coded admissions in children 0-3 y, Spain	<i>Epidemiologia</i>	Ecological panel	PM _{2.5} / PM ₁₀	Spain
Sun et al.	Fine particulate matter exposure and long-term lung-function trajectory in adults with cystic fibrosis	<i>Ann Am Thorac Soc</i>	Retrospective cohort	PM _{2.5}	USA
Wang	Environmental and socioeconomic influences on childhood asthma and pneumonia burden	<i>Front Med</i>	Retrospective cohort	PM _{2.5}	China
Yang	Allergenic vs. toxic risk of airborne seed hairs compared with ambient PM _{2.5}	<i>J Hazard Mater</i>	Field measurement	PM _{2.5} + bioaerosol	China
Yue et al.	Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou	<i>Front Public Health</i>	Time-series	Traffic gases (NO ₂ , SO ₂)	China
Zhang	Air pollution exposure in infancy and incidence of atopic dermatitis in preschool-aged children	<i>Epidemiology</i>	Prospective cohort	PM _{2.5} only	USA
Zhu et al.	Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis	<i>Medicine (Baltimore)</i>	Time-series	PM _{2.5} + PM ₁₀ + 4 gases	China
Mechanistic toxicology					
Balu et al.	SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction	<i>J Biochem Mol Toxicol</i>	Ex vivo perfused organ	Diesel PM (SRM 1650b)	India
Ding	PM _{2.5} promotes vascular calcification via miR-27b-5p/HES1 signalling	<i>J Environ Sci (China)</i>	In vitro + organoid + murine	PM _{2.5}	China
Easton et al.	Ultrafine particulate matter emitted from ships drives inflammation and susceptibility to viral infection	<i>Environ Int</i>	In vitro + organoid + murine	Ultrafine (ship-source)	Europe
Fakfum	Seasonal PM _{2.5} exposure and plasma metabolome changes related to metabolic syndrome	<i>Toxics</i>	Repeated-measures biomarker	PM _{2.5}	Thailand
Liu	Resveratrol ameliorates PM _{2.5} -aggravated IBS-D via gut bacteria and hypoxanthine/linoleic acid metabolism	<i>Environ Toxicol Pharmacol</i>	Murine inhalation exposure	PM _{2.5} only	China
Lu	Toxicity of PM _{2.5} from incomplete solid-fuel burning in <i>Caenorhabditis elegans</i>	<i>Toxics</i>	Controlled exposure (ecotox)	PM _{2.5} (biomass)	China
Maraschi	Osmoregulatory disruption and metal bioaccumulation from settleable atmospheric PM	<i>J Comp Physiol B</i>	Controlled exposure (ecotox)	Settleable APM	Brazil
Pan et al.	Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis	<i>Front Pharmacol</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀ / DEP (animal)	Global
Ribble et al.	Exposure timing is a determinant of fine particulate matter pulmonary, vascular, and metabolic toxicity in male mice	<i>Toxicol Sci</i>	Murine inhalation exposure	PM _{2.5}	USA
Wang	PM exposure, mitochondrial stress and extracellular matrix remodelling in scleral fibroblasts	<i>Exp Eye Res</i>	In vitro + organoid + murine	PM (unspecified) / emissions	China
Williams et al.	Liver-derived, circulating xanthine oxidoreductase drives vascular impairment associated with inhalation of ultrafine particulates	<i>Free Radic Biol Med</i>	Murine inhalation exposure	Ultrafine	USA
Wu	Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation	<i>J Environ Sci (China)</i>	Real-ambient murine + co-culture	Real-ambient PM	China
Zeng et al.	Circular RNA circDNAJC5 regulates ultrafine particle-disrupted microglial lipid metabolism via sphingolipid signalling	<i>Environ Health (Wash)</i>	In vitro (cell lines)	Ultrafine (PM0.1)	China
Occupational & indoor					
Bolden et al.	Wildfire smoke indoors: a review of infiltration risk factors and exposure mitigation strategies	<i>J Air Waste Manag Assoc</i>	Scoping review	Wildfire PM _{2.5} (indoor)	USA
Chen	Nested suction instrument to cut intraoperative surgical-smoke PM _{2.5}	<i>Front Med Technol</i>	Bench / device experiment	PM _{2.5} (surgical smoke)	China

First author	Focus	Journal	Design	Metric	Region
Cole	Determinants of respirable crystalline silica exposure in Australian tunnelling	<i>Ann Work Expo Health</i>	Observational exposure study	Respirable dust / RCS	Australia
Ding	Stage-resolved hair Zn-isotope dynamics as a firefighting combustion-exposure tracer	<i>J Hazard Mater</i>	Repeated-measures biomarker	Combustion-derived PM	China
Ghosh	Respiratory quality of life and pulmonary function among traffic police in eastern India	<i>Indian J Occup Environ Med</i>	Cross-sectional (multistage)	PM (unspecified) / emissions	India
Gonzalez-Colom et al.	Should household air quality monitoring be considered in selected patients with asthma and COPD?	<i>BMJ Open Respir Res</i>	Prospective cohort	PM _{2.5} (indoor)	Spain
Gwak	Size-resolved modelling of airborne viral RNA copies in indoor environments	<i>Risk Anal</i>	Physical exposure model	Bioaerosol / coarse	Global
Komonnithiphong	PM _{2.5} -bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants	<i>Toxics</i>	Field measurement	PM _{2.5} components	Thailand
Park et al.	Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon	<i>Sci Total Environ</i>	Field measurement	PM (indoor, unspecified)	South Korea
Qi	Emission rates of respirable crystalline silica and dust from engineered stone fabrication	<i>Ann Work Expo Health</i>	Bench / device experiment	Settleable / respirable dust	Global
Ramsden	Xenobiotic hazards in aircraft cabin air	<i>J Xenobiot</i>	Review + risk economics	Ultrafine particles	Global
Rogula-Kopiec et al.	Underground versus outdoor air in the Wieliczka Salt Mine health resort: particulate matter and selected gaseous pollutants	<i>Environ Geochem Health</i>	Field measurement	Size-resolved PM (PM ₁ -PM _{tot})	Poland
Subway PSD	Characterisation and source apportionment of PM and CO ₂ in subway stations with differing platform door forms	<i>J Environ Sci (China)</i>	Field measurement	PM _{2.5} / PM ₁₀	China
Sensing, forecasting & instrumentation					
Acker et al.	Comparison of policy-relevant air quality metrics calculated with sparse in situ monitoring and contiguous satellite-derived data	<i>Geohealth</i>	Satellite retrieval evaluation	PM _{2.5}	USA
Aher	Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy	<i>Environ Sci Pollut Res</i>	Commentary / methods letter	PM (unspecified)	India
Ainembabazi	Trends in atmospheric pollution research in Uganda, 1990-2025: a scoping review	<i>Toxics</i>	Scoping review	PM _{2.5} / PM ₁₀	Uganda
Ali et al.	Evaluation of low-cost sensors for size-resolved indoor particle monitoring	<i>Sci Rep</i>	Chamber sensor evaluation	PM _{2.5} + PM ₁₀ (size-resolved)	India
Bang et al.	Application of stationary IoT sensor networks to reduce exposure to particulate matter during local hotspot events in a megacity	<i>Sensors (Basel)</i>	Real-time sensor campaign	PM _{2.5} + PM ₁₀	South Korea
Campbell et al.	Comparison of black carbon measurements using filter-specific reference transmittance to lab blanks or averaged unloaded filters	<i>J Air Waste Manag Assoc</i>	Sensor co-location	Black carbon (filter eBC)	Global
Feng	Calibration strategies and measurement frameworks for low-cost particulate matter sensors: a review	<i>Meas Sci Technol</i>	Critical review	PM _{2.5} only	Global
Gharibzadeh et al.	Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor	<i>J Environ Health Sci Eng</i>	Sensor co-location	PM _{2.5} + PM ₁₀	Iran
Ginsburg	Global and regional calibration of low-cost PM _{2.5} sensors (PurpleAir): transferability of US-based corrections	<i>Aerosol Sci Technol</i>	Field measurement	PM _{2.5} only	Global
Holloway et al.	Applying satellite-derived PM _{2.5} data to policy-relevant air quality metrics	<i>Geohealth</i>	Satellite retrieval evaluation	PM _{2.5}	USA
Huda	Low-cost sensor network for wildfire monitoring and air quality at remote community locations	<i>J Air Waste Manag Assoc</i>	Field measurement	PM _{2.5} only	Canada
Jung et al.	Development of a high-resolution PM _{2.5} model in Taiwan using integrated low-cost air quality sensors and satellite observations	<i>J Expo Sci Environ Epidemiol</i>	Machine-learning model	PM _{2.5}	Taiwan
Kos Sinhala et al.	Determinants of in-home personal exposure to black carbon: the MobiSense sensor-based study	<i>Int J Environ Health Res</i>	Real-time sensor campaign	Black carbon	France
Msigwa	AIoT real-time sensing and calibration of low-cost PM sensors using the TCM network	<i>IEEE Internet Things J</i>	Deep-learning model	PM _{2.5} only	Global
Olivares Lopez et al.	Sensitivity of ultrafine particle and black carbon simulations to model resolution and emission data in urban environments	<i>Environ Pollut</i>	Chemical transport model	Ultrafine (PN) + black carbon	Spain

First author	Focus	Journal	Design	Metric	Region
Patel	Scalable air-quality sensor placement via gradient-based mutual information maximization	<i>Proc AAAI</i>	Deep-learning model	PM _{2.5} only	Global
Patton	Human health risk of particle exposure in outdoor environments in southwestern Pennsylvania	<i>Environ Sci Pollut Res</i>	Real-time sensor campaign	PM _{2.5} only	USA
Ryu et al.	Spatial variability and pollutant-specific mitigation effects of an urban vegetated park observed with a high-density air quality sensor network	<i>Environ Pollut</i>	Real-time sensor campaign	PM _{2.5} + gases	South Korea
Shi	Adaptive spatiotemporal graph transformer for multi-site multi-step PM _{2.5} forecasting	<i>J Environ Manage</i>	Deep-learning model	PM _{2.5}	China / India
Singh et al.	A spatiotemporal machine learning framework for high-resolution PM _{2.5} estimation using reconstructed satellite aerosol observations	<i>Environ Sci Pollut Res Int</i>	Machine-learning model	PM _{2.5}	India
Tsuchiya	Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices	<i>Biosensors</i>	Real-time sensor campaign	Bioaerosol / coarse	Japan
Wang	Joint monitoring and early warning of SARS-CoV-2 in outdoor PM _{2.5} and wastewater	<i>Viruses</i>	Environmental surveillance campaign	PM _{2.5} (bioaerosol)	China
Yu et al.	Integrating audiovisual data for short-term particulate matter concentration prediction on urban sidewalks	<i>Environ Sci Technol</i>	Machine-learning model	PM _{2.5} + PM ₁	China
Exposure assessment & modelling					
Chen	Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities	<i>J Hazard Mater</i>	Source apportionment	PM _{2.5}	China
Chen et al.	Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability	<i>J Hazard Mater</i>	Exposome-wide association	PM _{2.5} constituents + VOCs	USA
Chengdu NO ₃	Gas-phase reactions accelerate nitrate formation in spring and early summer in a Sichuan Basin megacity	<i>J Environ Sci (China)</i>	Supersite observation	PM _{2.5} components	China
Christoforou et al.	Ascertaining household-level exposure to air pollution in people with schizophrenia and other psychotic disorders	<i>Int J Popul Data Sci</i>	Registry cohort + geospatial	PM _{2.5} + PM ₁₀ + NO _x	UK
Ding	Urban-rural balanced estimation framework for PM _{2.5} exposure assessment across China	<i>Geo-spat Inf Sci</i>	Deep-learning model	PM _{2.5} only	China
Firooz et al.	Characterization of traffic-related particulate matter in an urban tunnel: source apportionment of exhaust and non-exhaust emissions	<i>Environ Pollut</i>	Source apportionment	PM _{2.5} + PM ₁₀	Iran
Hu	Fusion of ground monitoring, satellite observation and chemical transport modelling for seamless hourly 1-km PM _{2.5}	<i>Geo-spat Inf Sci</i>	Deep-learning model	PM _{2.5} only	China
Nguyen	Satellite-driven prediction of PM _{2.5} : machine learning and explainable AI	<i>Eng Res Express</i>	Critical review	PM _{2.5} only	Global
Nyoni et al.	Performance of road-traffic-based exposure proxies against personal PM _{2.5} measurements in three Sub-Saharan African countries	<i>Air Qual Atmos Health</i>	Proxy validation vs personal exposure	PM _{2.5} (personal, filtered)	The Gambia / Kenya / Mozambique
Pajak	Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk	<i>Toxics</i>	Dispersion model + soil sampling	PM ₁₀ / PAH	Poland
Patwary	Spatio-temporal assessment of PM ₁₀ and meteorological correlates in Cumilla, Bangladesh	<i>Res Sq (preprint)</i>	Field measurement	PM ₁₀	Bangladesh
Raj	Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh	<i>Res Sq (preprint)</i>	Field measurement	PM _{2.5} + PM ₁₀	India
Sanchez	Morbidity related to level-two eruption of Popocatepetl in communities near the crater	<i>Indian J Occup Environ Med</i>	Ecological panel	PM _{2.5} + PM ₁₀ jointly	Mexico
Solorzano-Araque	Multiscale exposure modelling: LOTOS-EUROS + mobility surveys + particle toxicity	<i>GeoHealth</i>	CTM + mobility integration	PM _{2.5} / PM ₁₀	Colombia
Su et al.	Industrial condensable particulate matter: a missing source in PM _{2.5} source apportionment driving winter pollution in China	<i>Environ Sci Technol</i>	Source apportionment	PM _{2.5}	China
Zhang	High-resolution aerosol liquid water content across the contiguous US using machine learning	<i>npj Clim Atmos Sci</i>	Machine-learning model	PM _{2.5}	USA

First author	Focus	Journal	Design	Metric	Region
Zhang et al.	Extrapolating personal exposure measurements to air pollution from outdoor sources into annual mean estimates	<i>Environ Int</i>	Physical exposure model	PM _{2.5} + BC + NO ₂ + O ₃	UK
Burden, policy & mitigation					
Abi Nader	Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict	<i>Disaster Med Public Health Prep</i>	Environmental surveillance campaign	PM _{2.5}	Lebanon
Cui et al.	Future chronic and acute air pollution deaths in China under climate change and emission reduction	<i>Environ Sci Technol</i>	Chemical transport model	PM _{2.5} + O ₃	China
Hu	Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants	<i>Front Public Health</i>	Bayesian + geospatial regression	PM ₁₀	China
Jiang et al.	Global disease burden attributable to ambient air pollution: disparities, determinants, and implications	<i>Health Data Sci</i>	GBD secondary analysis	PM _{2.5} + O ₃	Global
Karamzad	Ischaemic heart disease burden in MENA 1990-2021; PM attributable fraction 28.2%	<i>Health Sci Rep</i>	GBD secondary analysis	Ambient PM	MENA
Kraus	Development and evaluation of a risk-based air quality index for Germany	<i>Int J Hyg Environ Health</i>	Bayesian + geospatial regression	PM _{2.5} + PM ₁₀ jointly	Germany
Li	Burden trends of Alzheimer disease and other dementias in China, 1990-2023	<i>Ann Acad Med Singap</i>	GBD secondary analysis	PM (unspecified)	China
Makkar	Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems	<i>Environ Sci Pollut Res</i>	Critical review	PM _{2.5}	India
McDuffie et al.	Meteorologically driven changes in future global air quality: physical and monetized impacts	<i>Environ Sci Technol</i>	Chemical transport model	PM _{2.5} + O ₃	Global
Qian et al.	Stronger actions are needed to advance PM _{2.5} mitigation and its health benefits under population aging in China	<i>J Hazard Mater</i>	Machine-learning model	PM _{2.5}	China
Rees	HEPA filtration to reduce respiratory infection in care homes	<i>PLoS One</i>	RCT process evaluation	Filtered PM	UK
Shan	Policy-driven emission reductions and post-pandemic rebound, high-resolution inventories	<i>J Environ Sci (China)</i>	Emission inventory	PM emissions	China
Shkembi et al.	Workplace exposures may mask wildfire smoke-related exposure inequities and mortality	<i>Environ Sci Technol Lett</i>	Ecological panel	Wildfire PM _{2.5}	USA
Spira-Cohen et al.	Changes in respiratory and cardiovascular morbidity air pollution risk and attributable burden in New York City	<i>Environ Epidemiol</i>	Time-series	PM _{2.5} + NO ₂ + O ₃	USA
Sun	Clarifying causal and translational inference in studies of ambient PM _{2.5} and lung cancer	<i>Br J Cancer</i>	Commentary / methods letter	PM _{2.5}	Global
Vellaiyan	Diesel NO _x emissions and public health: source-pathway-receptor synthesis	<i>J Environ Health Sci Eng</i>	Narrative review	PM (unspecified)	Global
Zhou	Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis	<i>Toxics</i>	Spatial econometric model	Ultrafine particles	China
Other clinical endpoints					
Dogan Karatas	Air pollution and sudden sensorineural hearing loss: nine-year retrospective study	<i>J Clin Med</i>	Retrospective clinical series	PM ₁₀ / PM _{2.5}	Turkiye
He	PM _{2.5} -bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator	<i>Toxics</i>	Cross-sectional (multistage)	PM _{2.5} components	China
Hevia-Ramos et al.	Long-term exposure to air pollution and chronic kidney disease incidence: the Danish Nurse Cohort	<i>J Expo Sci Environ Epidemiol</i>	Prospective cohort	PM _{2.5} + PM ₁₀ + BC + NO ₂	Denmark
Ion-Andrei	Chronic air pollution exposure and the immunomolecular profile of lung cancer	<i>Biomedicines</i>	Retrospective cohort	PM ₁₀ (with NO ₂ /NO _x)	Romania
Knapova et al.	Daily relationship between air pollution and physical activity in an industrial region	<i>Appl Psychol Health Well Being</i>	Panel study	PM ₁₀	Czech Republic
Lupita	PM _{2.5} , preventive-care capacity and caries/edentulism burden across the EU	<i>Dent J</i>	Ecological panel	PM _{2.5}	EU-27
Ouyang	Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities	<i>J Hazard Mater</i>	Time-series	PM _{2.5} + PM ₁₀	China
Park	Long-term PM exposure and risk of androgenetic alopecia	<i>JAAD Int</i>	Nationwide cohort	PM _{2.5} / PM ₁₀	South Korea

First author	Focus	Journal	Design	Metric	Region
Sun	Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes	<i>J Neurosurg Pediatr</i>	Registry cohort + geospatial	PM _{2.5} (satellite-derived)	USA
Viau et al.	Traffic-related air pollution and the risk of invasive and borderline epithelial ovarian tumours	<i>Cancer Epidemiol Biomarkers Prev</i>	Case-control	NO ₂ (land-use regression)	Canada
Yuan et al.	Proteomic signatures of ambient air pollution and risk of type 2 diabetes	<i>Ecotoxicol Environ Saf</i>	Repeated-measures biomarker	PM _{2.5} + NO ₂ + SO ₂ + benzene	UK
Zeng et al.c	Ambient air pollution and risk of active tuberculosis: a population-based cohort study in Taiwan	<i>Int J Epidemiol</i>	Prospective cohort	PM _{2.5}	Taiwan
Zhou et al.	Environmental exposome study for prostate cancer in elderly men from Western China	<i>BMJ Open</i>	Case-control	PM _{2.5} + SO ₂	China
Zurita	Childhood leukaemia and proximity to benzene emission sources in the Mexico City area	<i>Pediatr Blood Cancer</i>	Ecological panel	PM (unspecified)	Mexico

Provenance and method — read this before citing any count above

This issue breaks the standing rollout contract, deliberately and on request. Rollups in this series normally aggregate previously published daily issues and never re-query an API. Daily coverage of July exists only for 27–31 July, so the 1–26 July portion of this corpus was **harvested independently on 1 August 2026** and had never appeared in any daily issue. Sources for that backfill: **PubMed** E-utilities on the [Date - Entry] field, 2026/07/01–2026/07/26, along two independent axes — health (352 records) and instrumentation (53) — giving 368 unique records after merge, of which 22 were already in `state/seen.json` from the daily issues and were dropped, 166 passed a PM-relevance title screen, and **51 were selected for full classification** on the basis of journal standing, methodological weight and subtopic coverage. Those 51 are written to `state/corpus/2026-07-26.json` with `backfill: true` and an `entry_window` recording the retrospective harvest.

The consequence for every count in this issue, stated plainly: the month's 157 records are **not** a complete census of July. They are 106 records from five exhaustively-screened daily windows plus 51 selected from 166 eligible records across the remaining 26 days. **The 1–26 July portion is therefore a curated sample, not a census, and subtopic proportions for that period are biased toward what was selected.** Specifically, sensing and exposure work was preferentially retained because it is the reader's primary field. The 23-record sensing count is a *floor*, not an estimate, and the correct reading of the headline finding is that sensing output in July was *at least* 23 records and certainly more — which strengthens rather than weakens the conclusion that the daily windows were under-recovering. No proportion in this issue should be quoted as a July prevalence.

The daily-trend panel is labelled a harvest diagnostic for the same reason: the 26 July bar is one batch, not one day. Records were deduplicated against `state/seen.json` on PMID, lowercased DOI and title signature before any summarisation, so no record appears twice across the daily issues and the backfill. Effect estimates were transcribed verbatim from abstracts and only where an interval was reported; 62 qualified. Subtopic, design, particle-metric and geography labels are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. Two records fetched without abstracts (*Am J Epidemiol* PMID 42429684; *J Am Coll Cardiol* PMID 42455106) were **excluded rather than summarised from their titles**, despite the JACC piece being directly on this issue's theme. DOIs carry through from PubMed's own record. Content derived from PubMed, Europe PMC and ClinicalTrials.gov metadata; abstracts remain the copyright of their respective publishers.